Practical Prescribed-Time Fault-Tolerant Tracking Control for Uncertain Nonlinear Systems via an Immediate Actuator Switching Strategy

Ben Niu, Yulong Ji, Huanqing Wang, Xinjun Wang, and Xudong Zhao

***Abstract*—This article addresses the practical prescribed-time fault-tolerant tracking control (PFTC) problem for a class of nonlinear systems characterized by uncertain control gains and external disturbances. A novel prescribed-time regulator (PTR) is proposed to construct the controller, thereby achieving the tracking control objective with arbitrary accuracy within the prescribed tracking time. The phenomenon of complete actuator faults is also considered, and a fault-tolerant control strategy with immediate actuator switching is designed to mitigate their impact. Furthermore, a self-regulating control gain based on an error-feedback mechanism is introduced to restrain the control input while improving the overall control performance. To overcome the issue of "complexity explosion" arising from the derivation of virtual controllers, a class of nonlinear filters is designed to estimate the values of the virtual controllers. The PFTC strategy proposed in this article ensures the following: 1) all the signals in the closed-loop system remain bounded; 2) the tracking error is practical prescribed-time tracking stable; and 3) after detection of actuator faults, the tracking error reconverges to the desired range within the prescribed recovery time. Lastly, a simulation example is provided to demonstrate the effectiveness of the proposed control strategy.**

***Index Terms*—Fault-tolerant control, prescribed-time regulator (PTR), prescribed-time tracking control, self-regulating control gain, uncertain nonlinear systems.**

I. Introduction

In RECENT years, the prescribed-time control theory has received significant attention for its fast convergence rate [1], [2], [3], [4], [5]. In fact, the prescribed-time control theory originated in the 1960s, when Peter proposed the theory of "short-time" stability for linear systems [6]. Unfortunately, this work did not receive much attention from researchers, and the PTC theory did not lead to significant development for a long period. With advances in Lyapunov stability theory and the development of control techniques, the PTC method has been revitalized. In early research, the prescribed-time control theory was primarily employed for the control of linear systems [7]. For instance, based on the output feedback technique, the prescribed-time control strategy was developed to address control problems in linear systems [8]. However, since most systems in practical applications are nonlinear, applying the prescribed-time control method to address the control problems of nonlinear systems is a challenging and significant issue [9], [10], [11], [12], [13]. In fact, several researchers have acknowledged this issue, and some effective control approaches have been proposed [14], [15], [16], [17]. For example, an adaptive prescribed-time control strategy was developed to handle time-varying uncertainties and time-delay disturbances. Although the results described above have solved some important theoretical problems, three issues remain in [18]. The first issue is that the nonlinear systems studied in existing research are relatively simple, and many types of uncertainties are not considered. The second issue is the phenomenon of "complexity explosion" in the controller design process [19], [20], [21], [22]. The final issue is the occurrence of actuator faults, including complete faults, in the controlled systems [23], [24], [25]. Therefore, it is of theoretical importance to study the prescribed-time control strategy while addressing the problem of "complexity explosion" for nonlinear systems with uncertainties and actuator faults.

Actually, the first issue can be effectively addressed by the adaptive control method in the backstepping design process [26], [27], [28], [29], [30], and the second issue can be handled by the nonlinear filter technique (NFT) [31], [32], [33]. By accurately estimating the value of intermediate virtual controllers, the NFT prevents the problem of "complexity explosion" [34], [35], [36]. Furthermore, the NFT has been combined with other control methods to produce many effective control approaches [37], [38], [39], [40]. Nevertheless, the existing NFT has not accomplished the control objectives within the prescribed time, which has restricted its application. Hence, there is great research value and urgency in integrating the NFT and the prescribed-time control method for uncertain nonlinear systems. In addition, research on actuator faults in controlled systems is receiving a lot of attention [41], [42], [43]. For example, a maximum power point tracking control system with adaptive

Received 17 February 2025; accepted 3 June 2025. Date of publication 24 June 2025; date of current version 20 August 2025. This work was supported by the National Key Research and Development Program under Grant 2022YFB4701302. This article was recommended by Associate Editor Z. Liu. *(Corresponding author: Yulong Ji.)*

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Color versions of one or more figures in this article are available at https://doi.org/10.1109/TSMC.2025.3577253.

Digital Object Identifier 10.1109/TSMC.2025.3577253

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active fault-tolerant capability was developed for wind power systems in [44]. The adaptive neural decentralized two-bit-triggered control problem for interconnected large-scale nonlinear systems in nonstrict feedback forms with actuator faults was studied in [45]. Although the aforementioned literature demonstrates practicability in solving many practical problems, the effectiveness of these control methods diminishes in the presence of complete actuator faults. Therefore, the study of how to address the problem of complete actuator faults is of significant theoretical importance.

In this study, an improved prescribed-time regulator (PTR)-based practical prescribed-time fault-tolerant tracking control (PFTC) strategy is proposed for the investigated uncertain nonlinear systems to achieve the prescribed-time tracking control objective. The following is a summary of the primary contributions of this article.

1) The tracking accuracy and the trajectory of the tracking error can be predefined by utilizing our proposed PTR, and the prescribed-time for achieving the tracking objective can be preselected and is not influenced by the system's initial conditions. Meanwhile, a type of nonlinear filters is combined with the PTC algorithm to solve the the problem of "complexity explosion," resulting in a significant reduction of the computational burden.
2) Unlike some existing controllers constructed within the framework of prescribed-time control strategies [46], [47], [48], which suffer from the singularity problem at the zero time point, our designed controller addresses this issue by adding an adjustable parameter to the PTR. In addition, a dynamic control gain based on an error-feedback mechanism is introduced to prevent the control input from oversizing while improving the tracking control performance.
3) Compared with many publications that only consider partial actuator faults [49], [50], [51], this article investigates the phenomenon of complete actuator faults and designs an event-triggered immediate actuator switching mechanism. This mechanism incorporates the prescribed-time control method, which ensures that for actuator faults occurring before the prescribed tracking time, the control performance is recovered before the prescribed tracking time, and for actuator faults occurring after the prescribed tracking time, the control performance is recovered before the prescribed recovery time.

II. Problem Statement and Preliminaries

Consider a class of uncertain nonlinear systems as follows:

$$\begin{cases} \dot{x}_i = g_i(\bar{x}_i)x_{i+1} + \theta_i^T\phi_i(\bar{x}_i) + b_i(t) \\ \dot{x}_n = g_n(\bar{x}_n)u + \theta_n^T\phi_n(\bar{x}_n) + b_n(t) \\ y = x_1 \end{cases} \tag{1}$$

where $i = 1, \ldots, n-1$, x_i, $\bar{x}_i = [x_1, x_2, \ldots, x_i]^T$, ϕ_i and u are the system states, the state vectors, the known functions and the control input, respectively. θ_i, g_i and b_i are the unknown vectors, the uncertain control gains and the external disturbances. y describes the system's output and y_r represents the desired trajectory.

Definition 1: Define the tracking error of the system (1) as $z_1 = x_1 - y_r$. If for any constants $\ell > 0$ and $T_f > 0$, the tracking error z_1 satisfies

$$\begin{aligned} &\|z_1(t)\| < \ell, \ t \geq T_f \\ &\|z_1(t)\| \leq b, \ 0 \leq t < T_f \end{aligned} \tag{2}$$

where $b > \|z_1(0)\|$ is a positive constant, then we call the tracking error is practical prescribed-time tracking stable.

Control Objective: This article focuses on designing a practical PFTC strategy for system (1) such that:

1) All the signals in the closed-loop system are bounded.
2) The tracking error of the nonlinear system (1) is practical prescribed-time tracking stable.
3) After detecting the occurrence of actuator faults, the tracking error will recover to the desired range within the prescribed-time.

Several assumptions are introduced to accomplish the control objective.

Assumption 1: The directions of $g_i(\cdot)$ are known and do not change. Moreover, there exist two unknown constants g_m and g_M such that

$$0 < g_m \leq |g_i(\cdot)| \leq g_M, \ i = 1, \ldots, n. \tag{3}$$

Remark 1: Assumption 1 indicates that $g_i(\cdot)$ are strictly positive or negative. Without loss of generality, we assume $g_i(\cdot) > 0$.

Assumption 2: The desired trajectory $y_r(t)$ and its derivatives $\dot{y}_r(t)$, $\ddot{y}_r(t)$ are bounded.

Assumption 3: For the external disturbances $b_i(t)$, there exist an upper bound $b_M > 0$ such that $|b_i(t)| \leq b_M$.

III. Main Results

A. Design of Dynamic Control Gain

Before presenting the design process of the prescribed-time controller, we begin by designing the following dynamic control gain based on the error-feedback mechanism:

$$H(\kappa) = \bar{c}\frac{e^{\sqrt{1+\kappa^2+c_0}}}{e^{\sqrt{1+\kappa^2+c_0}} - e} \tag{4}$$

where $\bar{c}$ and $0 < c_0 < 1$ are the positive design parameters and κ is the error variable. For any κ, the function $H(\kappa)$ has the following properties:

1) For any $|\kappa_1| \leq |\kappa_2|$, one can easily conclude that $H(\kappa_1) \leq H(\kappa_2)$, which indicates that the value of the function of dynamic gain increases when the absolute value of the variable decreases.
2) When the variable κ approaches to zero, $\lim_{\kappa \to 0+} H(\kappa) = 1/\sqrt{c_0}$, which implies that we can choose a smaller design parameter c_0 to obtain a larger control gain.
3) As the variable κ tends to positive infinity, the function $H(\kappa)$ converges to a constant, which indicates $\lim_{\kappa \to +\infty} H(\kappa) = c$.

Remark 2: It should be noted that our proposed dynamic control gain is a negative feedback regulator. Compared to

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using the constant gains, our proposed dynamic gains gradually converge to a constant when the error is excessively large, thus avoiding excessive control inputs, while the control gains are dynamically amplified when the error is small, thus improving the control performance.

B. Design of Prescribed-Time Regulator

The following type of PTR is established to accomplish the controller design:

$$\beta^*(t) = \begin{cases} \varphi(t)+\ell, & 0 \le t < T_f \\ \ell, & T_f \le t \end{cases} \tag{5}$$

where ℓ denotes the predefined tracking accuracy and T_f describes the prescribed time, $\varphi(t)$ is the crucial part of $\beta(t)^*$ and is selected such that

1) $\varphi(t)$ is continuous and n-order differentiable on $[0, T_f]$.
2) $\varphi(t)$ is monotonically increasing or decreasing on $[0, T_f]$.
3) $\varphi^{(1)}(T_f) = \varphi^{(2)}(T_f) = \cdots = \varphi^{(n)}(T_f) = 0, \varphi(0) = \gamma - \ell$,

where γ represents the initial error scope. In fact, there are many functions that satisfy the above definition of $\varphi(t)$, such as $[\gamma - \ell/T_f^k](T_f - t)^k$ $(k \ge n)$, $(\gamma-\ell)e^{(1/T_f^2)-[1/(T_f-t)^2]}$, etc. In this article, we select $[\gamma - \ell/T_f^n](T_f - t)^n$ with a simpler structure. Further, we can obtain the PTR $\beta(t)$ used in the controller design as

$$\beta(t) = \begin{cases} \frac{\gamma-\ell}{T_f^n}\left(T_f - t\right)^n + \ell, & 0 \le t < T_f \\ \ell, & T_f \le t. \end{cases} \tag{6}$$

Remark 3: It should be noted that some efficient PTRs have been proposed in the existing results under the frame of prescribed-time control [46], [47], [48], however, the values of these PTRs at zero time point tend to infinity, which cause the controllers designed with these PTRs to have the singular problem at the zero time point. By adding an adjustable parameter γ, our proposed PTR $\beta(t)$ avoids the singular problem mentioned above.

Define the adaptive parameters and the error transformation as

$$\delta_i = \left\{\|\theta_i^T\theta_i\| : i = 1, \ldots, n\right\}, \ \varepsilon_1 = \frac{z_1}{\beta(t)^2 - z_1^2}. \tag{7}$$

Apparently, δ_i is an unknown positive constant and will be estimated by $\hat{\delta}_i$. Further, for $|z_1(0)| < |\beta(0)|$, $\varepsilon_1(t)$ is bounded only if $|z_1(t)| < |\beta(t)|$ $(t > 0)$. Therefore, the tracking control objective is transformed to construct a PFTC strategy to ensure that ε_1 is bounded.

C. Design of Prescribed-Time Tracking Controller

For system (1), the prescribed-time controller design process includes n-steps. At first, to ensure the stability of the first i subsystems, the virtual controllers α_i are designed in steps $i(1 \le i \le n-1)$. Furthermore, the nonlinear filters are constructed to estimate the virtual controller's value in each step. In the end, the control input u will be designed in step n. In order to simplify the expression, the variables x_i, $\bar{x}_i$, and t will be omitted from the expression of the formulas during the controller design process.

Step 1: Select the positive definite function V_1 as

$$V_1 = \frac{1}{2}\varepsilon_1{}^2 + \frac{g_m}{2}\tilde{\delta}_1^2. \tag{8}$$

Differentiating V_1 yields

$$\dot{V}_1 = \varepsilon_1\dot{\varepsilon}_1 - g_m\tilde{\delta}_1\dot{\hat{\delta}}_1. \tag{9}$$

Combining the definition of ε_1 with the PTR (6) gives

$$\begin{aligned}\dot{\varepsilon}_1 &= \frac{\dot{z}_1}{\beta^2 - z_1^2} + \frac{2z_1^2}{\left(\beta^2 - z_1^2\right)^2}\dot{z}_1 - \frac{2z_1\beta}{\left(\beta^2 - z_1^2\right)^2}\dot{\beta} \\ &= \varrho\dot{z}_1 + \varpi\end{aligned} \tag{10}$$

where $\varrho = 1/(\beta^2 - z_1^2) + 2z_1^2/(\beta^2 - z_1^2)^2$ and $\varpi = -2z_1\beta\dot{\beta}/(\beta^2 - z_1^2)^2$. Consequently, by substituting the formula (10) into (9), one obtains

$$\dot{V}_1 = \varepsilon_1\varrho(\theta_1\phi_1 + g_1x_2 + b_1 - \dot{y}_r) + \varepsilon_1\varpi - g_m\tilde{\delta}_1\dot{\hat{\delta}}_1. \tag{11}$$

Then, applying the Young's inequality yields

$$\varepsilon_1\varrho\theta_1\phi_1 \le \varepsilon_1^2\varrho^2\theta_1^T\theta_1\phi_1^T\phi_1 g_m + \frac{1}{4g_m} \tag{12a}$$

$$\varepsilon_1\varrho b_1 \le \varepsilon_1^2\varrho^2 g_m + \frac{b_M}{4g_m} \tag{12b}$$

$$\varepsilon_1\varrho\dot{y}_r \le \varepsilon_1^2\varrho^2\dot{y}_r^2 g_m + \frac{1}{4g_m} \tag{12c}$$

$$\varepsilon_1\varpi \le \varepsilon_1^2\varpi^2 g_m + \frac{1}{4g_m}. \tag{12d}$$

Combining (11) with the inequalities (12), the following result is obtained:

$$\begin{aligned}\dot{V}_1 \le{}& \varepsilon_1\varrho\left[\varepsilon_1\varrho g_m\left(\delta_1\phi_1^T\phi_1 + 1 + \dot{y}_r^2\right) + g_1\alpha_1\right] - g_m\tilde{\delta}_1\dot{\hat{\delta}}_1 \\ &+ \varepsilon_1\varrho g_1(x_2 - \alpha_1) + \varepsilon_1^2\varpi^2 g_m + \zeta_1\end{aligned} \tag{13}$$

where $\zeta_1 = (3 + b_M)/4g_m$.

Subsequently, the virtual controller α_1 is constructed as

$$\alpha_1 = -k_1\frac{1}{\varrho}\varepsilon_1 - \frac{1}{\varrho}\varpi^2\varepsilon_1 - \varrho\left(1 + \dot{y}_r^2 + \hat{\delta}_1\phi_1^T\phi_1\right)\varepsilon_1 \tag{14}$$

where $k_1 = H(\varepsilon_1) + \bar{k}_1$, $\bar{k}_1$ is a non-negative design parameter.

Then, construct the update law of $\hat{\delta}_1$ as

$$\dot{\hat{\delta}}_1 = \varrho^2\varepsilon_1^2\phi_1^T\phi_1 - q_1\hat{\delta}_1 \tag{15}$$

where q_1 is the positive design parameter. Then, by combining the definition of $\hat{\delta}_1$ with (15), one can conclude $\hat{\delta}_1 > 0$.

Further, we can obtain the following conclusion by utilizing the previous result:

$$-g_1\varrho^2\varepsilon_1^2\hat{\delta}_1\phi_1^T\phi_1 \le -g_m\varrho^2\varepsilon_1^2\hat{\delta}_1\phi_1^T\phi_1. \tag{16}$$

Substituting (14)–(16) into (13), one can derive

$$\begin{aligned}\dot{V}_1 \le{}& -k_1\varepsilon_1^2 + \varepsilon_1\varrho g_1(x_2 - \alpha_1) + (g_m - g_1)\varepsilon_1^2\varrho^2\left(1 + \dot{y}_r^2\right) \\ &+ \varepsilon_1^2\varpi^2(g_m - g_1) + \zeta_1 + g_m q_1\tilde{\delta}_1\hat{\delta}_1.\end{aligned} \tag{17}$$

Since $0 < g_m < g_1$, the following inequality holds:

$$\dot{V}_1 \le -k_1\varepsilon_1^2 + g_m q_1\tilde{\delta}_1\hat{\delta}_1 + \varepsilon_1\varrho g_1(x_2 - \alpha_1) + \zeta_1. \tag{18}$$

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Step i ($2 \le i < n$): To handle the problem of "complexity explosion" of the virtual controller's derivatives, the nonlinear filters are designed in this step. Define the coordinate transformation of the ith subsystem as $\varepsilon_i = x_i - s_{i-1}$ and the boundary error as $e_{i-1} = s_{i-1} - \alpha_{i-1}$, where s_{i-1} represents the output of the nonlinear filter and satisfies $s_{i-1}(0) = \alpha_{i-1}(0)$. Now, construct the nonlinear filter as

$$\dot{s}_{i-1} = -\hat{M}_{i-1}^2 e_{i-1} - l_{i-1} e_{i-1} - \varepsilon_{i-1}^2 e_{i-1} \tag{19}$$

where l_{i-1} is a positive design parameter and M_{i-1} denotes the upper bound of the derivative of α_{i-1}, which will be estimated by $\hat{M}_{i-1}$.

Then, we select the following positive definite function:

$$V_i = V_{i-1} + \frac{1}{2}\varepsilon_i{}^2 + \frac{1}{2}e_{i-1}^2 + \frac{1}{2}\tilde{M}_{i-1}^2 + g_m \frac{1}{2}\tilde{\delta}_i^2. \tag{20}$$

Further, the derivative of V_i is calculated as

$$\begin{aligned}\dot{V}_i = {} & \dot{V}_{i-1} + \varepsilon_i\left(\theta_i^T\phi_i + g_i\alpha_i + b_i - \dot{s}_{i-1}\right) + g_i(\varepsilon_{i+1} + e_i) \\ & + e_{i-1}(\dot{s}_{i-1} - \dot{\alpha}_{i-1}) - \tilde{M}_{i-1}\dot{\hat{M}}_{i-1} - g_m\tilde{\delta}_i\dot{\hat{\delta}}_i.\end{aligned} \tag{21}$$

Since $g_m > 0$, the following inequalities obviously hold:

$$\varepsilon_i\theta_i\phi_i \le \varepsilon_i^2\theta_i^T\theta_i\phi_i^T\phi_i g_m + \frac{1}{4g_m} \tag{22a}$$

$$\varepsilon_i b_i \le \varepsilon_i^2 g_m + \frac{b_M}{4g_m} \tag{22b}$$

$$\varepsilon_i\dot{s}_{i-1} \le \varepsilon_i^2\dot{s}_{i-1}^2 g_m + \frac{1}{4g_m} \tag{22c}$$

$$g_{i-1}\varepsilon_{i-1}\varepsilon_i \le \varepsilon_{i-1}^2\varepsilon_i^2 g_m + \frac{g_M}{4g_m} \tag{22d}$$

$$g_{i-1}\varepsilon_{i-1}e_{i-1} \le \varepsilon_{i-1}^2 e_{i-1}^2 + \frac{g_M}{4}. \tag{22e}$$

Define the constant ζ_i as

$$\zeta_i = \frac{1}{4} + \frac{1}{2g_m} + \frac{b_M}{4g_m} + \frac{g_M}{4g_m} + \frac{g_M}{4}. \tag{23}$$

By substituting (22) and (23) into (21), one gives

$$\begin{aligned}\dot{V}_i \le {} & -\sum_{j=1}^{i-1} k_j\varepsilon_j^2 + \sum_{j=1}^{i}\zeta_j + \sum_{j=1}^{i-1} g_m q_j\tilde{\delta}_j\hat{\delta}_j + g_i(x_{i+1} + e_i) \\ & + \varepsilon_i\left[g_m\varepsilon_i\left(\delta_i\phi_i^T\phi_i + 1 + \dot{s}_{i-1}^2 + \varepsilon_{i-1}^2\right) + g_i\alpha_i\right] \\ & + e_{i-1}(\dot{s}_{i-1} - \dot{\alpha}_{i-1}) - \tilde{M}_{i-1}\dot{\hat{M}}_{i-1} - g_m\tilde{\delta}_i\dot{\hat{\delta}}_i \\ & + \sum_{j=1}^{i-2}\tilde{M}_j\hat{M}_j + \varepsilon_{i-1}^2 e_{i-1}^2 - \frac{1}{4}.\end{aligned} \tag{24}$$

Then, by utilizing Young's inequality, we can can conclude

$$\begin{aligned}& e_{i-1}(\dot{s}_{i-1} - \dot{\alpha}_{i-1}) = e_{i-1}\dot{s}_{i-1} - e_{i-1}\dot{\alpha}_{i-1} \\ & = -\hat{M}_{i-1}^2 e_{i-1}^2 - l_{i-1}e_{i-1}^2 - \varepsilon_{i-1}^2 e_{i-1}^2 - e_{i-1}\dot{\alpha}_{i-1} \\ & \le -\hat{M}_{i-1}^2 e_{i-1}^2 - l_{i-1}e_{i-1}^2 - \varepsilon_{i-1}^2 e_{i-1}^2 + |e_{i-1}|M_{i-1}.\end{aligned} \tag{25}$$

Consequently, we can derive the following result:

$$\begin{aligned}|e_{i-1}|M_{i-1} &= \tilde{M}_{i-1}|e_{i-1}| + \hat{M}_{i-1}|e_{i-1}| \\ &\le \tilde{M}_{i-1}|e_{i-1}| + \hat{M}_{i-1}^2 e_{i-1}^2 + \frac{1}{4}.\end{aligned} \tag{26}$$

By substituting (25) and (26) into (24), one yields

$$\begin{aligned}\dot{V}_i \le {} & -\sum_{j=1}^{i-1} k_j\varepsilon_j^2 - \sum_{j=1}^{i-1} l_j e_j^2 + \sum_{j=1}^{i}\zeta_j + \sum_{j=1}^{i-1} g_m q_j\tilde{\delta}_j\hat{\delta}_j \\ & + \varepsilon_i\left[g_m\varepsilon_i\left(\delta_i\phi_i^T\phi_i + 1 + \dot{s}_{i-1}^2 + \varepsilon_{i-1}^2\right) + g_i\alpha_i\right] \\ & + \tilde{M}_{i-1}\left(|e_{i-1}| - \dot{\hat{M}}_{i-1}\right) - g_m\tilde{\delta}_i\dot{\hat{\delta}}_i \\ & + g_2\varepsilon_i(\varepsilon_{i+1} + e_i) + \sum_{j=1}^{i-2} p_j\tilde{M}_j\hat{M}_j.\end{aligned} \tag{27}$$

The intermediate virtual controller α_i and the adaptive laws $\dot{\hat{M}}_{i-1}$, $\dot{\hat{\delta}}_i$ are designed as

$$\begin{aligned}\alpha_i &= -k_i\varepsilon_i - \varepsilon_i\left(\hat{\delta}_i\phi_i^T\phi_i + 1 + \dot{s}_{i-1}^2 + \varepsilon_{i-1}^2\right) \\ \dot{\hat{M}}_{i-1} &= |e_{i-1}| - p_{i-1}\hat{M}_{i-1},\ \ \dot{\hat{\delta}}_i = \varepsilon_i^2\phi_i^2 - q_i\hat{\delta}_i.\end{aligned} \tag{28}$$

where $k_i = H(\varepsilon_i) + \bar{k}_i$, $\bar{k}_i$ is a non-negative design parameter, p_{i-1} and q_i are the positive design parameters. Now, combining (27) with (28), one can obtain

$$\begin{aligned}\dot{V}_i \le {} & -\sum_{j=1}^{i} k_j\varepsilon_j^2 - \sum_{j=1}^{i-1} l_j e_j^2 + \sum_{j=1}^{i-1} p_j\tilde{M}_j\hat{M}_j + \sum_{j=1}^{i} g_m q_j\tilde{\delta}_j\hat{\delta}_j \\ & + (g_m - g_i)\varepsilon_i^2\left(\delta_i\phi_i^T\phi_i + 1 + \dot{s}_{i-1}^2 + \varepsilon_{i-1}^2\right) \\ & + \sum_{j=1}^{i}\zeta_j + g_i\varepsilon_i(\varepsilon_{i+1} + e_i).\end{aligned} \tag{29}$$

Since $0 < g_m < g_1$, the following conclusion holds:

$$\begin{aligned}\dot{V}_i \le {} & -\sum_{j=1}^{i} k_j\varepsilon_j^2 - \sum_{j=1}^{i-1} l_j e_j^2 + \sum_{j=1}^{i-1} p_j\tilde{M}_j\hat{M}_j + \sum_{j=1}^{i} g_m q_j\tilde{\delta}_j\hat{\delta}_j \\ & + \sum_{j=1}^{i}\zeta_j + g_i\varepsilon_i(\varepsilon_{i+1} + e_i).\end{aligned} \tag{30}$$

Step n: Define the coordinate transformation of the nth subsystem as $\varepsilon_n = x_n - s_{n-1}$ and the boundary error as $e_{n-1} = s_{n-1} - \alpha_{n-1}$. Then, design the nonlinear filter as

$$\dot{s}_{n-1} = -\hat{M}_{n-1}^2 e_{n-1} - l_n e_{n-1} - \varepsilon_{n-1}^2 e_{n-1}. \tag{31}$$

Choose the positive definite function V_n as

$$V_n = V_{n-1} + \frac{1}{2}\varepsilon_n{}^2 + \frac{1}{2}e_{n-1}^2 + \frac{1}{2}\tilde{M}_{n-1}^2 + g_m\frac{1}{2}\tilde{\delta}_n^2 \tag{32}$$

where M_{n-1} denotes the upper bound of the derivative of α_{n-1}, $\hat{M}_{n-1}$ is the estimation of M_{n-1} and $\tilde{M}_{n-1} = M_{n-1} - \hat{M}_{n-1}$ represents the estimate error. Then, the derivative of V_n is described as

$$\begin{aligned}\dot{V}_n = {} & \dot{V}_{n-1} + \varepsilon_n\left(\theta_n^T\phi_n + g_n u + b_n - \dot{s}_{n-1}\right) - g_m\tilde{\delta}_n\dot{\hat{\delta}}_n \\ & + e_{n-1}(\dot{s}_{n-1} - \dot{\alpha}_{n-1}) - \tilde{M}_{n-1}\dot{\hat{M}}_{n-1}.\end{aligned} \tag{33}$$

Now, design the control input u, the adaptive laws $\dot{\hat{M}}_{n-1}$ and $\dot{\hat{\delta}}_n$ as

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$$u = -k_n\varepsilon_n - \varepsilon_n\left(\hat{\delta}_n\phi_n^T\phi_n + 1 + \dot{s}_{n-1}^2 + \varepsilon_{n-1}^2\right)$$
$$\dot{\hat{M}}_{n-1} = |e_{n-1}| - p_{n-1}\hat{M}_{n-1},\ \ \dot{\hat{\delta}}_n = \varepsilon_n^2\phi_n^2 - q_{n-1}\hat{\delta}_n. \quad (34)$$

where $k_n = H(\varepsilon_n) + \bar{k}_n$, $\bar{k}_n$ is a non-negative design parameter, p_{n-1} and q_n are the positive design parameters.

Since $0 < g_m < g_n$, the following conclusion is valid by substituting (31) and (34) into (33):

$$\dot{V}_n \le -\sum_{j=1}^{n} k_j\varepsilon_j^2 - \sum_{j=1}^{n-1} l_j e_j^2 + \sum_{j=1}^{n-1} p_j\tilde{M}_j\hat{M}_j + \sum_{j=1}^{n} g_m q_j\tilde{\delta}_j\hat{\delta}_j + \sum_{j=1}^{n}\zeta_j. \quad (35)$$

D. Stability Analysis

After completing the controller design, we will analyze the stability of the closed-loop system.

Theorem 1: Consider the nonlinear system (1) under Assumptions 1–3 without actuator faults, there exists a set of control signals defined by (14), (28), and (34) such that

1) All the the signals in the closed-loop system are bounded.
2) The tracking error z_1 is practical prescribed-time tracking stable.
3) The system error variable $\varepsilon = [\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n]^T$ will converge to the specified compact set by adjusting the values of $\bar{c}$ and c_0 in the dynamic gain.

Proof: 1) First, we select the Lyapunov function as $V{=}V_n$ for the stability analysis. Based on the previous conclusion, the derivative of V is described as (35). Further, for the terms $\tilde{M}_j\hat{M}_j$ and $\tilde{\delta}_j\hat{\delta}_j$, we can conclude

$$\sum_{j=1}^{n-1} p_j\tilde{M}_j\hat{M}_j \le \sum_{j=1}^{n-1}\frac{1}{2}M_j^2 - \sum_{j=1}^{n-1} p_j^2\frac{1}{2}\tilde{M}_j^2$$
$$\sum_{j=1}^{n} q_j\tilde{\delta}_j\hat{\delta}_j \le \sum_{j=1}^{n}\frac{1}{2}\delta_j^2 - \sum_{j=1}^{n} q_j^2\frac{1}{2}\tilde{\delta}_j^2. \quad (36)$$

By combining (35) with (36), one gives

$$\dot{V} \le -\sum_{j=1}^{n}\left(k_j\varepsilon_j^2 + g_m q_j^2\tilde{\delta}_j^2\right) - \sum_{j=1}^{n-1}\left(l_j e_{n-1}^2 + p_j^2\tilde{M}_j^2\right) + \lambda \le -cV + \lambda \quad (37)$$

where $\lambda = \sum_{j=1}^{n}\zeta_j + \sum_{j=1}^{n-1}M_j^2 + \sum_{j=1}^{n}\delta_j^2$ stands for the extra constant, $c = 2\min\{\bar{c} + \bar{k}_j, l_j, p_j^2, q_j^2\}$. Consequently, by integrating the inequality (37), we can obtain

$$V \le \left(V(0) - \frac{\lambda}{c}\right)e^{-ct} + \frac{\lambda}{c},\ \ t \ge 0 \quad (38)$$

which means for any $t \ge 0$, the error vectors $\varepsilon = [\varepsilon_1, \ldots, \varepsilon_n]^T$, $e = [e_1, \ldots, e_{n-1}]^T$, $\tilde{\delta} = [\tilde{\delta}_1, \ldots, \tilde{\delta}_n]^T$, $\tilde{M} = [\tilde{M}_1, \ldots, \tilde{M}_{n-1}]^T$ belongs to the following compact set:

$$\Omega = \left\{\left[\varepsilon, e, \tilde{\delta}, \tilde{M}\right] | V\left(\left[\varepsilon, e, \tilde{\delta}, \tilde{M}\right]\right) \le V(0) + \frac{\lambda}{c}\right\}. \quad (39)$$

Hence, ε_i, $\tilde{\delta}_i$ $(i = 1, \ldots, n)$, e_r, and $\tilde{M}_r$ $(r = 1, \ldots, n-1)$ are bounded. Since δ_i and M_r are unknown positive constants, we can obtain that $\hat{\delta}_i$ and $\hat{M}_r$ are bounded. Consequently, it's easy to verify α_i are also bounded. Moreover, based on the definitions of ε_i and the boundary error e_r, it can be concluded that x_i is bounded, indicating that all the signals are bounded.

2) Based on (7), one can deduce that ε_1 is bounded only if $|z_1| < |\beta(t)|$. From the above proof, ε_1 is bounded for all $t \ge 0$. Therefore, we can obtain that $|z_1| < |\beta(t)|$, which indicates that z_1 is always limited in the desired scope $\beta(t)$. According to the definition of $\beta(t)$, one obtains $\beta(t) = \ell$ for $t \ge T_f$, where ℓ is the predefined accuracy and T_f is the prescribed time. Since $|z_1| < |\beta(t)|$ is true for $t \ge T_f$, we can conclude that the tracking error z_1 of the system (1) is reduced to the predefined boundary ℓ within the prescribed tracking time T_f.

3) Now, we are going to render the positive effect of the introduction of dynamic gain on the stability of the controlled system.

Based on (37), one can deduce

$$\begin{aligned}\dot{V} &\le -\sum_{j=1}^{n} k_j\varepsilon_j^2 + \lambda \le -\sum_{j=1}^{n} H(\varepsilon_j)\varepsilon_j^2 + \lambda \\ &\le -\sum_{j=1}^{n}\bar{c}\frac{e^{\sqrt{1+\varepsilon_j^2+c_0}}}{e^{\sqrt{1+\varepsilon_j^2+c_0}} - e}\varepsilon_j^2 + \lambda \\ &\le -\bar{c}\frac{e^{\sqrt{1+V_s+c_0}}}{e^{\sqrt{1+V_s+c_0}} - e}\sum_{j=1}^{n}\varepsilon_j^2 + \lambda \\ &\le -\bar{c}\frac{e^{\sqrt{1+V_s+c_0}}}{e^{\sqrt{1+V_s+c_0}} - e}V_s + \lambda \\ &\le -\bar{c}\frac{e^{V_s+c_0}}{e^{V_s+c_0} - 1}V_s + \lambda \end{aligned} \quad (40)$$

where $V_s = \sum_{j=1}^{n}\varepsilon_j^2$ denotes the aggregate potential energy of the system errors. Further, define the following nonlinear function

$$F_s(V_s) = -\bar{c}\frac{e^{V_s+c_0}}{e^{V_s+c_0} - 1}V_s + \lambda. \quad (41)$$

Then, we can calculate the derivate of F_s, which is described as

$$F_s'(V_s) = \bar{c}\left(\frac{e^{V_s+c_0}}{\left(e^{V_s+c_0} - 1\right)^2}V_s - \frac{e^{V_s+c_0}}{e^{V_s+c_0} - 1}\right) \quad (42)$$

Obviously, the derivate of F_s is less than zero.

Subsequently, we are able to deduce that V_s will asymptotically converge to a new compact set, which is described as

$$\Omega_s = \left\{V_s | \bar{c}\frac{e^{V_s+c_0}}{e^{V_s+c_0} - 1}V_s \ge \lambda\right\} \quad (43)$$

which means that we can adjust the parameter c_0 to converge the error variable of the closed-loop system to an arbitrarily compact set.

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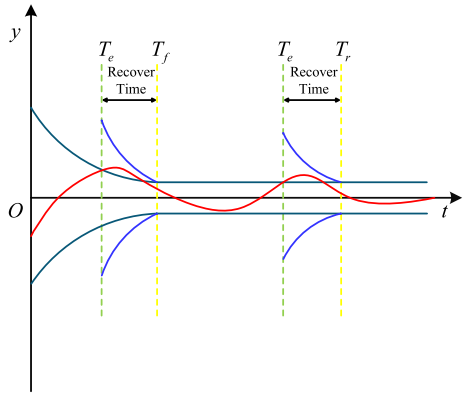


Fig. 1. Fault-tolerant control model based on the immediate switching mechanism.

Remark 4: Although the constant gain can drive the system error to converge to a specified compact set, it must be chosen as an extremely large initial value. However, when the system error is large, the control input can become infinite, potentially destabilizing the system. In contrast, the dynamic gain proposed in this article is based on error-feedback, which drives the system error to asymptotically converge while continuously amplifying the control gain. This allows the error to asymptotically converge to an arbitrarily compact set.

E. Fault-Tolerant Control Strategy

For the problem of actuator fault, this article proposes a fault-tolerant control strategy based on the immediate switching mechanism of the actuator. If the tracking error reaches a predefined boundary, which indicates that the actuator is faulty and cannot be recovered automatically, the actuator will be switched automatically in this case. Subsequently, we assume that the time at which the systematic error reaches a predefined boundary is T_e, and the prescribed recovery time T_r is fixed as $T_r = T_f - T_e$ when $T_e < T_f$ to ensure that the prescribed-time control objective can be achieved. Similarly, when $T_e \geq T_f$, it demonstrates that the prescribed-time control objective has been achieved, at this case the prescribed recovery time is predefined as T_r.

Consequently, according to the previous analysis, we first define the prescribed recover time regulator (PRTR) before the prescribed tracking time as

$$\beta_1(t) = \begin{cases} \frac{\gamma_1 - \ell}{(T_f - T_e)^n}(T_f - t)^n + \ell, & T_e \leq t < T_f \\ \ell, & T_f \leq t. \end{cases} \tag{44}$$

where γ_1 is the predefined recovery boundary. Subsequently, the PRTR after the prescribed tracking time is constructed as

$$\beta_2(t) = \begin{cases} \frac{\gamma_1 - \ell}{T_r^n}(T_r + T_e - t)^n + \ell, & T_e \leq t < T_r + T_e \\ \ell, & T_f \leq t. \end{cases} \tag{45}$$

Based on the designed PRTR, we can obtain the the graphical representation of the tracking performance as below.

As we can see from Fig. 1, the red line represents the output of the considered nonlinear system, the black line represents the constraint boundaries of the initial actuator, and the blue line represents the regulator boundaries of the alternate actuator. The green dotted line denotes the time point when the system detects an actuator faults, at which time the system will automatically switch to the alternate actuator.

Therefore, the switching condition for the actuator is designed as

$$|z_1| = |\beta(t)| \tag{46}$$

where $\beta(t)$ represents the PTR in the initial actuator. Meanwhile, utilizing the conclusions obtained during the backstepping design process in the previous section, we can derive all the control signals in the alternate controller as follows:

$$\begin{cases} \varrho = \frac{1}{\beta^2 - z_1^2} + \frac{2z_1^2}{(\beta^2 - z_1^2)^2}, \varpi = -\frac{2z_1\beta\dot{\beta}}{(\beta^2 - z_1^2)^2} \\ \bar{\varepsilon}_1 = \frac{z_1}{\bar{\beta}(t)^2 - z_1^2}, \varepsilon_i = x_i - s_{i-1}\varepsilon_n = x_n - s_{n-1}, \\ e_{i-1} = s_{i-1} - \alpha_{i-1}, e_{n-1} = s_{n-1} - \alpha_{n-1} \\ \dot{\hat{\delta}}_1 = \varrho^2\bar{\varepsilon}_1^2\phi_1^T\phi_1 - q_1\hat{\delta}_1, \dot{\hat{\delta}}_i = \varepsilon_i^2\phi_i^2 - q_i\hat{\delta}_i \\ \dot{\hat{\delta}}_n = \varepsilon_n^2\phi_n^2 - q_{n-1}\hat{\delta}_n, \dot{\hat{M}}_{i-1} = |e_{i-1}| - p_{i-1}\hat{M}_{i-1} \\ \dot{\hat{M}}_{n-1} = |e_{n-1}| - p_{n-1}\hat{M}_{n-1} \\ \dot{s}_{i-1} = -\hat{M}_{i-1}^2 e_{i-1} - l_{i-1}e_{i-1} - \varepsilon_{i-1}^2 e_{i-1} \\ \dot{s}_{n-1} = -\hat{M}_{n-1}^2 e_{n-1} - l_n e_{n-1} - \varepsilon_{n-1}^2 e_{n-1} \\ \alpha_1 = -k_1\frac{1}{\varrho}\bar{\varepsilon}_1 - \frac{1}{\varrho}\varpi^2\bar{\varepsilon}_1 - \varrho\left(1 + \dot{y}_r^2 + \hat{\delta}_1\phi_1^T\phi_1\right)\bar{\varepsilon}_1 \\ \alpha_i = -k_i\varepsilon_i - \varepsilon_i\left(\hat{\delta}_i\phi_i^T\phi_i + 1 + \dot{s}_{i-1}^2 + \varepsilon_{i-1}^2\right) \\ u = -k_n\varepsilon_n - \varepsilon_n\left(\hat{\delta}_n\phi_n^T\phi_n + 1 + \dot{s}_{n-1}^2 + \varepsilon_{n-1}^2\right) \end{cases} \tag{47}$$

where $2 \leq i < n$ and $\bar{\beta}$ is a recovery PTR satisfies $\bar{\beta} = \beta_1$ for $T_e < T_f$, $\bar{\beta} = \beta_2$ for $T_f \leq T_e$.

Theorem 2: Assuming that the actuator failed and the tracking error reaches the predefined boundary, the control signal in (47) can be utilized to ensure that the tracking error z_1 reconverges to the prescribed accuracy within the prescribed recovery time.

Proof: First, we select the Lyapunov function as

$$\bar{V} = \frac{1}{2}\bar{\varepsilon}_1^2 + \frac{1}{2}\sum_{j=2}^{n}\varepsilon_j^2 + \frac{1}{2}\sum_{j=1}^{n-1}e_j^2 + \frac{1}{2}\sum_{j=1}^{n-1}\tilde{M}_j^2 + \frac{1}{2}\sum_{j=1}^{n}g_m\tilde{\delta}_j^2 + \frac{1}{2}\sum_{j=1}^{n-1}s_j^2. \tag{48}$$

By substituting (47) into (48), it's easy to conclude that

$$\begin{aligned} \dot{\bar{V}} &\leq -k_1\bar{\varepsilon}_j^2 - \sum_{j=1}^{n-1}k_j\varepsilon_j^2 - \sum_{j=1}^{n}g_m q_j^2\tilde{\delta}_j^2 \\ &\quad - \sum_{j=1}^{n-1}\left(l_j e_{n-1}^2 + p_j^2\tilde{M}_j^2\right) + \lambda \\ &\leq -cV + \lambda \end{aligned} \tag{49}$$

where $\lambda = \sum_{j=1}^{n}\zeta_j + \sum_{j=1}^{n-1}M_j^2 + \sum_{j=1}^{n}\delta_j^2$ stands for the extra constant and $c = 2\min\{k_j, l_j, p_j^2, q_j^2\}$. Therefore, it is easy to validate that the fault-tolerant control strategy designed in this article guarantees that the tracking error can be recovered to the specified accuracy within the prescribed recover time T_r.

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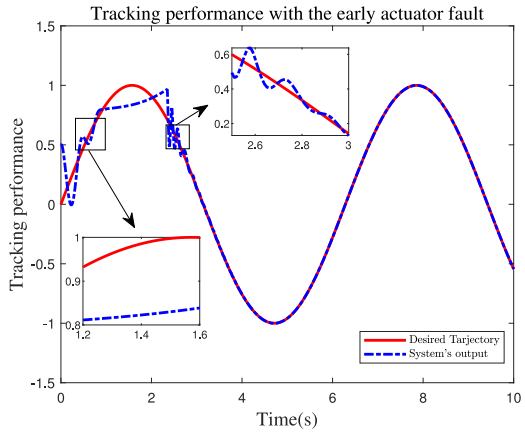


Fig. 2. Tracking performance when the actuator fault occurs at 1 s.

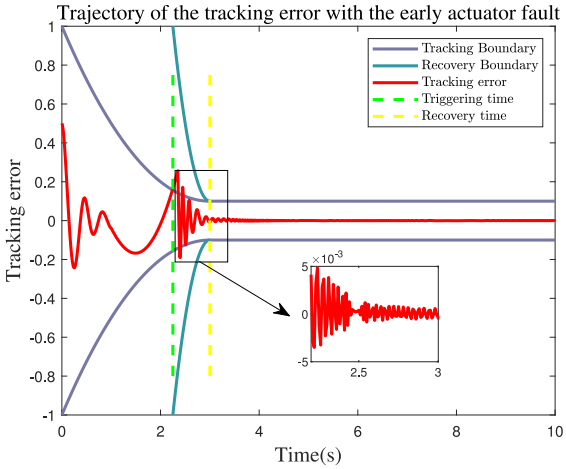


Fig. 3. Trajectory of the tracking error z_1 when the actuator fault occurs at 1 s.

Remark 5: In this article, a fault-tolerant control strategy is developed by combining the immediate actuator switching technique with the proposed PRTR. In contrast to existing fault-tolerant control methods, we construct separate recovery functions for different occurrence times of actuator faults, which not only ensures that the system accomplishes the prescribed time tracking objective, but also guarantees that the system reconverges to the specified accuracy after the actuator fault occurs.

IV. Simulation Example

This section provides a simulation example to examine the effectiveness of the proposed PFTC method. Consider the following uncertain nonlinear system:

$$\begin{cases} \dot{x}_1 = \frac{1}{4}x_1^3 + \frac{1}{2}\cos(t^2) + (\sin^2(x_1)+1)x_2 + \frac{1}{10}\cos^2(t) \\ \dot{x}_2 = \frac{3}{10}x_1x_2 + \frac{1}{5}\sin(x_2^3) + (\cos^2(x_1)+1)u + \frac{1}{10}\sin^2(t) \\ y = x_1 \end{cases} \tag{50}$$

where $\theta_1 = [(1/4),(1/2)]^T$, $\theta_2 = [(3/10),(1/5)]^T$, $\phi_1 = [x_1^3, \cos(t^2)]$, $\phi_2 = [x_1x_2, \sin(x_2^3)]$, $g_1 = \sin^2(x_1)+1$, $g_2 = \cos^2(x_1)+1$, $b_1 = (1/10)\cos^2(t)$ and $b_2 = (1/10)\sin^2(t)$.

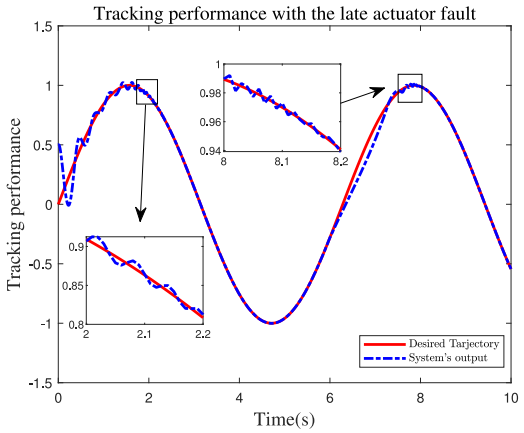


Fig. 4. Tracking performance when the actuator fault occurs at 6 s.

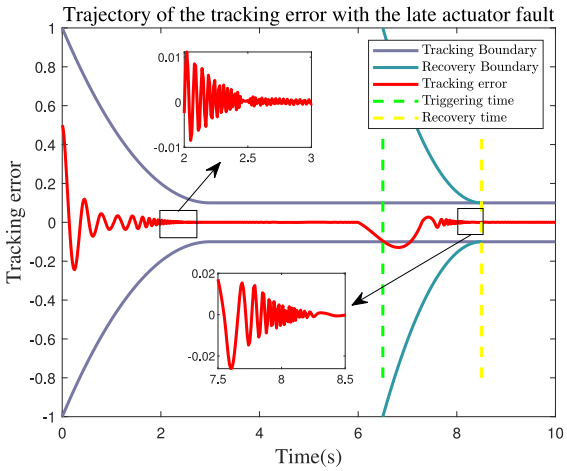


Fig. 5. Trajectory of the tracking error z_1 when the actuator fault occurs at 6 s.

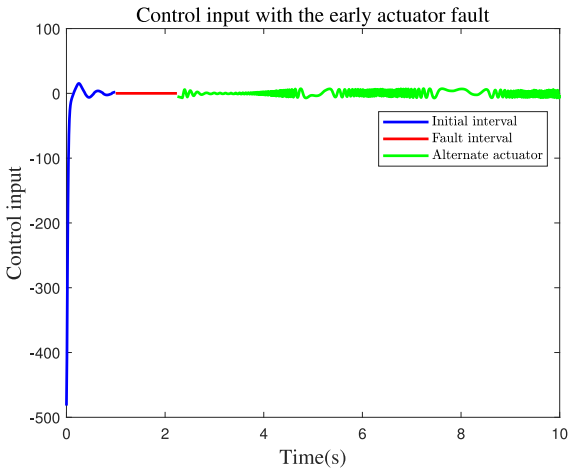


Fig. 6. Control input u when the actuator fault occurs at 1 s.

The system's initial condition is chosen as $[x_1(0), x_2(0)] = [0.5, 0]$. The design parameters are selected as $k_1 = 3$, $k_2 = 3$, $\gamma = 1$, $\gamma_1 = 1$, $q_1 = 1$, $q_2 = 1$, $p_1 = 0.5$, $\bar{c} = 1$, $c_0 = 0.01$, $[\hat{\delta}_1(0), \hat{\delta}_2(0), \hat{M}_0(0)] = [1, 1, 1]$ and $s_1(0) = \alpha_1(0)$. The reference signal, the prescribed tracking time T_f, the prescribed recovery time T_r and the predefined accuracy

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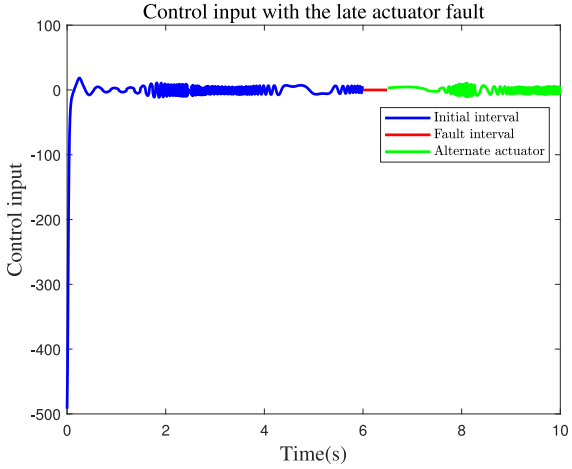


Fig. 7. Control input u when the actuator fault occurs at 6 s.

ℓ are selected as $T_f = 3$ s, $T_r = 2$ s, $\ell = 0.1$, respectively. We assume that the actuator is completely faulty and the fault time are assigned as 1 s and 6 s, respectively.

As shown in Fig. 2 and Fig. 3, when the actuator fault occurs before the prescribed tracking time T_f, our proposed control strategy guarantees that the tracking objective is achieved within T_f. As illustrated in Fig. 4 and Fig. 5, when the actuator fault occurs after the prescribed tracking time T_f, our proposed control strategy ensures that the tracking error z_1 converges to the predefined accuracy ℓ within the prescribed recovery time T_r. The control inputs of the nonlinear system are presented in Fig. 6 and Fig. 7, which indicates that both the original and alternate controllers are bounded.

V. Conclusion

In this article, the PFTC problem for the uncertain nonlinear system has been studied. A class of PTRs have been developed to ensure accurate tracking within the prescribed tracking time. Meanwhile, the nonlinear filters have been used to estimate virtual controllers, thus overcoming the "complexity explosion" phenomenon. To improve control performance while avoiding transient overloads, the self-regulating control gain has been designed to construct the controller. In addition, the fault-tolerant control strategy with immediate actuator switching method have been proposed to handle issue of the actuator faults. In the end of this article, the simulation example has been provided to demonstrate the effectiveness of the proposed control strategy.

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